

**Multiplying and Dividing in Scientific Notation**Question Bank - Additional 4.02 **ANSWER KEY***Show your work. Circle your final answer.***For questions 1 through 6, multiply or divide. Write each answer in scientific notation.**

1.  $(3 \times 10^2)(3 \times 10^6)$

$9 \times 10^8$

2.  $\frac{6.3 \times 10^7}{2.1 \times 10^3}$

$3 \times 10^4$

3.  $(4 \times 10^5)(8 \times 10^6)$

$32 \times 10^{11} = 3.2 \times 10^{12}$

4.  $\frac{2.4 \times 10^4}{8 \times 10^7}$

$0.3 \times 10^{-3} = 3 \times 10^{-4}$

5.  $(2.5 \times 10^{-5})(6 \times 10^{-4})$

$15 \times 10^{-9} = 1.5 \times 10^{-8}$

6.  $\frac{9.9 \times 10^{-3}}{3.3 \times 10^{-8}}$

$3 \times 10^5$

$-3 - (-8) = 5$

**Real quantities.**7. The Amazon River discharges about  $5.5 \times 10^7$  gallons of water every second.a. About how many gallons does it discharge in one hour, which is  $3.6 \times 10^3$  seconds?**about  $1.98 \times 10^{11}$  gallons** $5.5 \cdot 3.6 = 19.8$ , so the product is  $19.8 \times 10^{10}$  before it is finished.b. A second river discharges about  $1.1 \times 10^7$  gallons per second. How many times larger is the Amazon's rate?**5 times larger**

8. A factory produces about  $6.4 \times 10^5$  bottles per day.

a. How many bottles does it produce in  $3.5 \times 10^2$  days?

$$2.24 \times 10^8 \text{ bottles}$$

$$6.4 \cdot 3.5 = 22.4 \text{ .}$$

b. A second factory produced  $7.2 \times 10^8$  bottles in  $1.2 \times 10^3$  days. How many bottles per day is that?

$$6 \times 10^5 \text{ bottles per day}$$

### Working backwards.

9. Find the missing number in each equation.

a.  $(4 \times 10^n)(2 \times 10^3) = 8 \times 10^{10}$

$$n = 7$$

$$n + 3 = 10 \text{ .}$$

b.  $\frac{7.2 \times 10^6}{n \times 10^2} = 2.4 \times 10^4$

$$n = 3$$

$$7.2 \div n = 2.4 \text{ .}$$

c.  $(2.5 \times 10^3)(8 \times 10^5) = \underline{\hspace{2cm}}$

$$2 \times 10^9$$

*The coefficients multiply to exactly 20.*

### Reading someone else's work.

10. Dana divides  $\frac{4.8 \times 10^{11}}{1.2 \times 10^4}$  and writes  $3.6 \times 10^7$ . Name her mistake and give the correct quotient.

*She subtracted the coefficients instead of dividing them. The power of ten is right. The correct quotient is  $4 \times 10^7$  .*

$$4.8 \div 1.2 = 3.6 \text{ , which is where her coefficient came from; } 4.8 \div 1.2 = 4 \text{ .}$$

11. Which is the value of  $\frac{9.6 \times 10^{-3}}{3.2 \times 10^5}$  ?

● A.  $3 \times 10^{-8}$

○ B.  $3 \times 10^8$

○ C.  $6.4 \times 10^{-8}$

○ D.  $3 \times 10^2$

**A**

12. Select ALL of the expressions whose value is  $8 \times 10^6$  .

☒ **A.**  $(1.6 \times 10^3)(5 \times 10^3)$

☒ **B.**  $\frac{1.6 \times 10^{10}}{2 \times 10^3}$

☐ **C.**  $(4 \times 10^2)(2 \times 10^8)$

☒ **D.**  $\frac{4 \times 10^8}{5 \times 10^1}$

☒ **E.**  $(4 \times 10^2)(2 \times 10^4)$

☐ **F.**  $\frac{1.6 \times 10^{10}}{2 \times 10^4}$

**A, B, D and E**

*B and D each give  $0.8 \times 10^7$  , which is  $8 \times 10^6$  once it is finished.*